LAB 12: AXON MODEL, A BIOPHYSICS LABORATORY

During this laboratory session, students will

✔1 do a biophysics laboratory about axon,

✔2 build and study three different axon models,

✔3 use knowledge from Ohm's law, in particular the resistance and capacitance addition rules,

✔4 use knowledge form RC circuit,

✔5 use a circuit board.

Student Learning Outcome: Successful students will

✔1 strengthen their knowledge in Ohm's law,

✔2 strengthen their knowledge in RC circuit,

✔ 3 get acquainted to different homemade models of axon.

INTRODUCTION

In this lab, we will build three different homemade models of axon and will try to understand some basic electrical functionalities of axons. We will make the first one out of AL foil and plastic wrap. This very simple model will allow us to study the capacitor of an axon (myelinated or not). We will use carbon resistances for the second model to build a simplified version of Hodgkin–Huxley model. In this second model, we will focus on the equivalent resistance of the axon. The third model includes capacitors; thus, it is more elaborated than the first two. We will use the third model to measure the speed of the electric signal along the axon.

You have already used all the equipment, so each one of you has already acquired skills that will allow you to accomplish task without much help. This lab might be done really quickly with an organized group work. The three different activities can be prepared (and done) in any order and each one of you may start one of them at any time of the lab.

Please feel free to add any goals that you want to the abovementioned ones and include them in your lab report. Also, start thinking about the project you want for physics lab as time will go fast—an extension of this lab is a candidate.

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LAB 12: PRELIMINARY WORK

How does the capacitance depend on the overlapping area and the distance between the two "plates"?

Capacitance is defined as $C = \varepsilon A/d$ for parallel plate capacitors where ε is vacuum permittivity, A is the area of the plates, and d is the distance between them.

Two capacitors C_1 and C_2 are mounted in parallel, how would the equivalent capacitance be compared to C_1 and C_2? Would the equivalent capacitor be bigger if C_1 and C_2 were mounted in series? Give argument(s) why the myelinated axons have lower capacitance than unmyelinated ones. You may use sketches and pictures.

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When two capacitors are connected in parallel, the equivalent capacitance is C_1 + C_2, which is larger than either individual capacitor. This is because placing them in parallel effectively increases the total plate surface area, and since capacitance is proportional to area (C = εA/d), the combined capacitance grows.

When connected in series, the equivalent capacitance is smaller than

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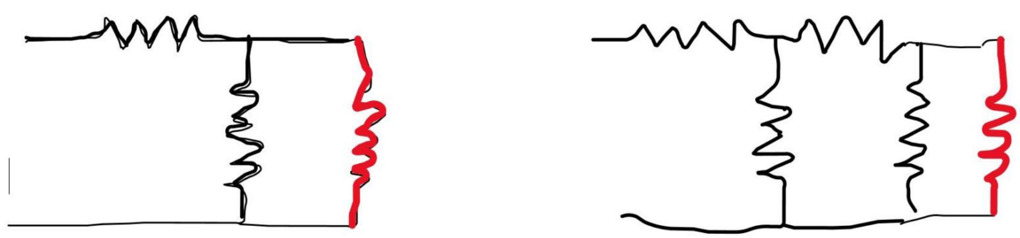
Calculate the equivalent resistances of each of the following two circuits. The resistances that are drawn horizontally are all 400 Ω, the resistances that drawn vertically in black are 1.5 kΩ, and the red ones are 1.0 kΩ.



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Circuit 1: R_eq = 1000 Ω

Circuit 2: R_eq = 2000 Ω

In the previous two figures, resistors that are drawn horizontally belong to the same group and those drawn vertically are part of another group. Which resistor(s) would represent the resistor(s) <u>along</u>

the axon in a slice and which one represent the resistor(s) of the axon <u>across</u> the membrane in the slice? (Just a common sense if you answer #3 correctly.)

The horizontal resistors represent the intracellular (axial) resistance and the vertical resistors represent the membrane resistance

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Consider the circuit on the left side in question #3. Assume that the potential difference between the two ends (that are free in the figure) is 3 V. Each resistance would drive some current, which you will calculate. Then you are asked to remove the pair of resistances (0.4 kΩ, 1.5 kΩ) so that the red one would be left alone. Would the electric current driven by the red resistance remain the same? Why or why not?

No, the current through the red resistor would not remain the same. When the 400 Ω and 1.5 kΩ resistors are present, the series 400 Ω resistor consumes part of the voltage (1.2 V), leaving only 1.8 V across the parallel pair, and the black 1.5 kΩ resistor shares the current load, so the red resistor only carries 1.8 mA. Once both are removed, the full 3 V is applied directly across the red resistor alone, increasing the current to 3.0 mA.

81 words

Consider the Fig. on the right side in question #3. You are asked to add another pair of resistances (0.4 kΩ and 1.5 kΩ) without changing the overall equivalent resistance. Draw your new circuit (Hint: See answer to question #3).

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In our third model of axon, we will omit the red resistance. However, each resistance in the ladder rung must be in parallel with the capacitor of the axon. Draw the circuit of our third model with at least four ladder rungs.

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Activity #1: Homemade axon and its capacitance (no more than 20 minutes)

This homemade axon consists of two capacitor plates made of two sheets of AL foil that are one insulated from the other by plastic wrap strips—in other words, the two Al sheets sandwich the plastic wrap. Two small Al strips will be used as plate connector leads or terminals.

Lay out one or more layers of plastic wrap on a table. Place one sheet of AL foil over the plastic wrap and place one of the terminal strips crossways near one end of it, half of it extending out from the sheet. Make sure that the plastic wrap is larger than the AL foil, yet the terminal extends beyond the plastic.

Lay the second layers of plastic wrap on top. Once again, be sure that the insulators are larger than the capacitor plates so the plates do not touch.

Cover this with the final sheet of AL foil, again placing it to cover what has already been laid down. Locate the second terminal extending in the direction opposite to the first terminal.

Starting at one end and working toward the other end, roll or fold this entire system into a roll—do not flatten nor roll it too tight. Make sure the two terminals are not connected (ohmmeter should indicate OL). You have now made the first model of axon, which is merely a capacitor!

Now, it is time to measure the capacitance of your axon with the capacitance meter. Include your result in your lab report. Make sure that you have your data before moving to the next step, **which all of you should observe together**.

Next step, while you have the axon hooked up, compress it and measure the new capacitance. Would you expect the capacitance to increase? Why or why not (preliminary work #1 may help)? Please include also this new value into your lab report (OL means Overload) and explain why a pinch, let us say on your hand, can hurt by comparing to the two measured values of the capacitance—remember that Q = CV.

Activity #2: (40 minutes) Question #4 can be done at home when writing the report

1. Build the circuit in the preliminary work #3, figure left, using PASCO equipment. If there would be no more 400 Ω, you must use four 100 Ω mounted in series to make up the 400 Ω (use a breadboard or the PASCO platform).

    a. Measure the equivalent resistance using a multimeter and include the result into your lab report.

    b. Fed the circuit with a 3 V DC battery (combination of two 1.5 V or output #1 of PASCO 850 interface). Measure the current driven by each resistance and the corresponding voltage drop. Report your results.

2. Do the same as in #1 for the circuit on the right side.
3. Again, do the same measurements with your circuit in preliminary work #6.
4. Answer the following questions.
    a. Do you expect a change in the currents? Why or why not? Support your answers with measured values and Kirchhoff's junction rule.
    b. Examine the ratios of voltage drops across two consecutive ladder rungs and compare them. Compare it also to 15/(15 + 10). Can you predict that ratio in function of the resistance values? You are expected to write the relationship giving the ratio in function of R_{across}, R_{long}, and the 1.0 kΩ resistor.

Activity #3: (About 80 minutes)

The charged particles responsible for electric currents driven by an axon are actually ions that are either moving down the body of the axon or across the membrane of the axon. The two electric currents, along the axon and across its membrane, exist simultaneously. As the ions moving along the axon remain inside the axon (of course), the resistors that are drawn horizontally in the figure in the preliminary work represent the axon body with resistance R_{long}. You will use the drawing in preliminary work #7 for this activity—you have to remove the 1.0 kΩ.

1. Draw the circuit for the third model with at least five ladder rungs (extension of preliminary work #7), including the labels for intracellular/inside fluid (cytosol) and extracellular/outside fluid.
2. Build the circuit you draw for #1 above. You will have either a set of 470 μF capacitors **or** a set of 47 μF. Do not forget to state the set of capacitors you have when writing your lab report. This question asks you to only build the circuit (no measurement yet).

3. Let us assume we apply a DC voltage that remains constant all the time (time-independent voltage). In that case, the voltage across the membrane depends on the position of the slice (or ladder rung) with respect to the first ladder rung. It is given by

$$V(n) = \varepsilon_o * e^{-n*L/\lambda}$$

"n" is the number of ladder rungs from the beginning (n = 1, 2, 3, 4, 5). "L" is the length of a slice—that is the measure of the length of each 400 Ω resistance. λ (lambda) is the so-called "length constant." According to the cable model, $\lambda/L = \text{sqrt}(R_{across}/R_{long})$. Find V(n) using the given values of your resistances (this is just calculations) if $\varepsilon_o = 3$ V.

The inside (R_{long}) should be connected to the positive plate of the battery to mimic an action potential. Be careful to connect the POLARIZED capacitors accordingly—the negative plate of the capacitor is indicated by the sign "—."

4. Steps #f, #g, and #h could be done at home, but do not forget to export your PASCO data table via Excel.

   a. Turn on PASCO 850 interface and connect it to your laptop. Double-click on Capstone. Connect the voltage sensor to the interface, then set it up in Capstone Hardware. Grab the "graph" and "table" into the display (*y*-axis = V, *x*-axis = time) and change the unit of time to milliseconds.

   b. We will use the output #1 of PASCO 850 interface: change it into DC, and set the voltage to 3 V. Connect the leads (red and black) to your circuit.

   c. Let us try to measure the voltage at the first slice. Thus, mount the voltage sensor in parallel to that slice.

   d. Change the frequency of measurement to at least 200 Hz—the default is 20 Hz, but you may increase it as much as needed. Now, click "record" the measurement in PASCO, then turn on the 6 V output power in the hardware system. Then turn it off as soon as the graph of the voltage stabilizes at its maximum. Export the data into Excel.

e. Mount the voltage sensor to the next slice. Then repeat #d.

f. Repeat #e for all the other slices. You should have exported the data for five graphs.

g. Choose a specific instantaneous time "t" (e.g., t = 0.32 seconds), then plot V(n), where "n" is the order of slice, that is to say n = 1 is the first slice, n = 2 is the second slice, and so on.

h. Use an exponential trendline to determine the experimental values of ε_o and λ.

i. The speed of the propagation of the signal is given by λ/τ. Find the experimental value of the speed—you may just use R_{across}*C for the time constant τ.

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